Project Design Phase-I
Problem – Solution Fit Template

Date	19 September 2022
Team ID	
Project Name	Health360: Executive Analytics & Population Health
Maximum Marks	Dashboard2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) — CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

- Target Audience: Healthcare executives and operational leads overseeing a 5,000-patient cohort.
- **Key Personas:**

1. **Dr. Michael Carter** (Chief Medical Officer / Clinical Lead) focusing on high-risk cardiovascular patients and vitals (BP vs. Cholesterol).
2. **Emily Rodriguez** (Director of Wellness Programs) managing lifestyle risk factors (Smoking, Alcohol, Physical Activity, BMI).
3. **Dr. Priya Nair** (Head of Population Health) monitoring demographic age brackets (Children to Elder), symptom prevalence, and resource planning.

6. CUSTOMER CONSTRAINTS — CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

- **Time Limitations:** Executive users have minimal time to manipulate raw data or build custom queries during strategy meetings.
- **Technical Barrier:** Need for an intuitive, zero-code, web-accessible interface without requiring local software installations.
- **Data Volume:** Processing and filtering 5,000 patient records without performance lag or slow query rendering.

5. AVAILABLE SOLUTIONS — AS

Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

- Legacy static Excel/CSV tabular spreadsheets requiring manual pivot tables.
- Monthly printed operational reports lacking interactive drill-down capabilities.
- Disjointed EHR (Electronic Health Record) views without macro-level cohort visuals.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

- **Core Job:** Identify high-risk patient cohorts early and allocate preventive clinical and wellness resources efficiently.
- **Key Problems:**

1. Inability to correlate clinical vitals with composite patient risk scores across static tabular reports.
2. Multi-variable lifestyle habit combinations (smoking + low exercise) remain obscured in isolated spreadsheets.
3. Slow manual aggregation of demographic age distributions and chronic disease prevalence.

9. PROBLEM ROOT CAUSE — RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

Healthcare data is trapped in raw relational or flat CSV formats (smart_healthcare_dataset.csv) without automated pre-calculated risk algorithms or interactive visual mapping layers.

7. BEHAVIOUR — BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits;
Indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

- **Executive stakeholders** interact with visual summary dashboards prior to clinical resource decisions.
- They continuously cross-examine age groups (**Children** through **Elder**) against health outcomes to isolate risk drivers.
- They use guided narratives (storyboards) to present actionable findings to healthcare committees.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS — TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

- Increasing rates of preventable chronic condition complications (Diabetes, Heart Disease, Stroke) across the cohort.
- Quarterly clinical quality reviews requiring rapid multi-factor risk stratification.
- Urgent need to reallocate wellness program budgets to high-risk behavioral groups.

4. EMOTIONS: BEFORE / AFTER — EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

- **Before:** Frustrated by slow, static data tables; anxious about missing critical cardiovascular warning signs; overwhelmed by fragmented lifestyle datasets.
- **After:** Empowered by dynamic cross-filtering; confident in data-driven clinical decision-making; clear on resource allocation priorities.

10. YOUR SOLUTION — SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

- **Health360 Tableau Master Dashboard & Executive Story:**

1. Unified single-screen dashboard featuring 9 interactive worksheets.
2. Real-time global cross-filtering across vitals scatter plots, lifestyle heatmaps, and demographic pyramids.
3. Pre-calculated dynamic measures (Age Category, Calculated Risk Score).
4. Guided 9-point Tableau Story providing step-by-step narrative insights for Dr. Carter, Emily Rodriguez, and Dr. Priya Nair.

8. CHANNELS of BEHAVIOUR — CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

Online: Cloud-hosted Tableau Public Server web links for real-time interactive browser exploration on desktop and tablets.

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

Offline / Embedded: Exported executive presentation decks and PDF summaries generated directly from Tableau Story points.

Extract online & offline CH of BE